

IS/IT Backsourcing decision making - A design science research approach

Jacky Akoka^a, Isabelle Comyn-Wattiau^{b,*}^a CEDRIC-CNAM, Paris, France^b ESSEC Business School, Cergy, France

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ABSTRACT

Many organizations implement outsourcing solutions for information systems and/or information technology (IS/IT). Some of them face problems resulting from dissatisfaction with these outsourcing decisions. Reasons for dissatisfaction include an outsourcing agreement that did not meet expectations, organizational changes, and a loss of control over the business. As a result, companies are rethinking their existing contracts by adopting a backsourcing solution. IS/IT backsourcing refers to the full or partial repatriation of previously outsourced activities. Given the strategic stakes of backsourcing, it is imperative to develop a decision-making process based on a solid framework. Based on an extensive literature review, we propose a design science research (DSR) approach consisting of four artifacts: a set of requirements, a conceptual framework, a decision model, and a prototype to support decision makers considering IS/IT backsourcing issues. A validation is presented, consisting of an illustrative scenario, a logical argument, and an expert evaluation. In addition, the approach was compared with recommendations found in the literature. Finally, the comments of a panel of IS/IT experts were collected. Our decision support system can be used by decision makers to investigate the backsourcing of IS/IT activities, choosing between three scenarios: backsource, renegotiate the contract, or change vendor.

1. Introduction

Sourcing is defined as “the act by which work is contracted or delegated to an external or internal entity that may be physically located anywhere” [1]. It refers to a make or buy decision, the outcome of which is insourcing or outsourcing. Information systems and information technology (IS/IT) insourcing is defined as the process that allows organizations to keep IS/IT activities within the organization. Conversely, IS/IT outsourcing involves the delegation of certain activities to an external third party, either domestically or offshore. IS/IT backsourcing is the full or partial repatriation of previously outsourced IS/IT activities by an organization, including the definitive termination or non-renewal of an existing outsourcing contract [2]. It can be defined as “a business practice in which an organization brings back in-house assets, activities, and capabilities that are part of its information systems operations and that were previously outsourced to one or more external IS providers” [3]. Hirschheim and Lacity define it as the re-internalization of a previously outsourced activity [4]. They were the first to coin the term backsourcing.

Although it is difficult to find recent figures on the extent of IS/IT

backsourcing, it is well known that many companies have used IS/IT backsourcing, including JPMorgan, Deutsche Telekom, Sainsbury's, Bouygues Telecom and others. It is easier to find statistics on outsourcing provided by companies that sell outsourcing services and promote them with eloquent figures on the outsourcing phenomenon, potentially masking that of backsourcing. However, without providing specific figures, the Deloitte Global Outsourcing Survey 2022 report¹ mentions a recent and significant phenomenon of companies insourcing their IS/IT services but locating them abroad (offshore insourcing). The ability of IS/IT staff to work remotely has facilitated the emergence of the Global in-house center (GIC) model. Faced with new challenges related to digital transformation, changing business needs and the opportunities presented by emerging technologies, organizations are being asked to rethink their past IS/IT sourcing decisions [5]. More and more companies are stepping up their digital transformation activities [6]. As a result, the needs of the stakeholders involved in these transformations are changing [7]. In addition, the range of technologies and solutions is growing [8]. As a result, organizations are rethinking their existing IS/IT sourcing arrangements. In addition, many organizations have to deal with the issues arising from dissatisfaction with the implemented

* Corresponding author at: ESSEC Business School, 3 Av. B. Hirsch, 95021 Cergy Cedex, France.

E-mail addresses: jacky.akoka@lecnam.net (J. Akoka), Isabelle.wattiau@essec.edu (I. Comyn-Wattiau).

¹ www2.deloitte.com/content/dam/Deloitte/us/Documents/process-and-operations/us-global-outsourcing-survey-2022.pdf (Accessed 2024, 30 Oct)

outsourcing solutions and the need to rethink and eventually adopt a back sourcing solution. Although companies are being asked to consider a multi-sourcing strategy, this paper will focus solely on back sourcing. In fact, back sourcing is one aspect of sourcing decisions [1]. It is a strategic decision that should consider the influence of the decision maker's beliefs. Factors that can influence this decision include improving IT service quality, relationship quality, regaining control over IS/IT, and reducing costs [5,9,10]. Given the financial, managerial, and strategic challenges of back sourcing, it is imperative to develop a decision-making process based on a sound framework.

Rajaeian et al. [11] compare 133 research articles on model-driven decision support for IT outsourcing and cloud sourcing. None are dedicated to back sourcing decisions. They conclude with a number of recommendations to improve the rigor and relevance of decision support systems research. They also mention the lack of practical approaches such as design science research (DSR), case study and action research. Veltri et al. propose a decision tree that identifies the problems and opportunities that can lead to a back sourcing decision [3]. However, this decision tree does not help to decide when two elements are contradictory. More generally, they state that the decision to back source is a complex one based on many criteria, as all their case studies attest. Therefore, we adopt a holistic view that facilitates decision making in complex environments. Drawing on several theories [12], we propose a DSR approach that includes four artifacts: a set of requirements, a conceptual framework, a decision model, and a prototype to support decision makers considering IS/IT back sourcing issues. Our DSR approach extends previous studies and can be used by decision makers to investigate the back sourcing of various activities.

The remainder of this paper is structured as follows. The second section reviews the current state of knowledge in back sourcing. The following section describes our research approach to facilitating back sourcing decisions, and its main steps. The fourth section describes our validation process. The last section discusses the managerial and theoretical contributions of the research, lists the main limitations, concludes, and discusses further research directions.

2. Current state of knowledge

The aim of this literature review is twofold: to describe previous contributions and to identify gaps to be filled, for which our approach proposes new artifacts. The issue of IS/IT back sourcing has already been the subject of several literature reviews, in particular one dealing with the back sourcing process [9], another focusing on the back sourcing of software development [13], and a third analyzing IS back sourcing in the wider landscape of IS outsourcing [14]. It is also worth mentioning the review of the literature on outsourcing by Dibbern et al. which is often cited in articles that deal with back sourcing [15]. Beyond these different literature reviews, the current state of knowledge covers four main topics: reasons and drivers for back sourcing; frameworks and models that analyze the back sourcing phenomenon; back sourcing process itself; and case studies from which best practices can be derived.

Many *reasons, motivations and drivers* lead companies to back source [3,16–18]. A categorization of drivers for back sourcing decisions into contract problems, internal and external opportunities is proposed by Veltri et al. [3]. Lack of cost savings and service degradation are the two main drivers identified by Hirshheim and Lacity in relation to the category of contract problems [16]. By distinguishing strategic factors from organizational factors, Wong and Jaya obtain four categories instead of the three proposed by Veltri et al. [17]. The strategic aspect is an important motivation according to Qu et al. who analyze a large number of cases of companies that have back sourced [18]. Felser and Wynn contrast the academic categorization of Veltri et al. with a practitioner's view that emphasizes cost savings, quality improvements and increased control and flexibility as the three most important reasons for back sourcing [5]. Thakur-Wernz uses these drivers to derive a typology of four types of back sourcing: profitability back sourcing, operational

back sourcing, strategic back sourcing and failure back sourcing [19]. Whitten and Leidner suggest that product quality, service quality, relationship quality and switching costs are related to the decision to back source [20]. Moreover, product and service quality are the differentiators in deciding whether a company will back source or switches vendors. Von Bary et al. found that only switching costs are likely to prevent back sourcing [10]. Finally, managerial and pragmatic concerns are relevant drivers in local government [21].

Several *frameworks or models* have been proposed: a framework that describes the mechanisms through which outsourcing failure affects back sourcing decisions [2], a framework that identifies how companies carry out the back sourcing process [22], and a framework based on task interdependency and task expertise to define four back sourcing scenarios [23]. Von Bary's 4-axis configuration framework allows the comparison of eight back sourcing cases reported in the literature [24]. Felser and Wynn propose a conceptual framework based on a knowledge-based view of the firm [5]. In terms of models, McLaughlin and Peppard propose a model based on patterns that summarize the issues that lead to a reassessment of the outsourcing decision [25]. Finally, Solli-Sæther and Gottschalk propose a stage-of-growth model for sourcing decisions [26].

Some studies focus on *understanding the process of back sourcing* [18,22,23,25,27], including its alternatives. The process of back sourcing is particularly efficient when associated with IT-enabled business processes, leading to superior firm performance [18]. The process of back sourcing reveals knowledge transfer and reintegration as a co-ordination activity, complementing the knowledge transfer mechanisms reported in the IS outsourcing literature [23]. McLaughlin and Peppard propose an end-to-end model describing the sourcing process in which back sourcing is an option [25]. Slyngstad examines how companies carry out the back sourcing process [22]. She identifies internal challenges that may arise during this process within a marine equipment supplier. Law addresses organizational disruption by examining the back sourcing process as an outsourcing disruption [27]. Alternatives to the back sourcing process include renewing, extending, or renegotiating the existing contract [25], switching vendors [20] and partial back sourcing [28].

Several studies synthesize qualitative research on IS/IT back sourcing *case studies and derive best practices*. Overby believes that back sourcing is more expensive than continuing to outsource, at least in the short term [29]. Butler et al. present the results of a case study carried out in an organization that had recently back sourced its entire IT department [30]. Nujen et al. analyze a company in the Norwegian shipbuilding industry, and present critical success factors for knowledge reintegration when bringing manufacturing back in-house [31]. Some key strategies can ensure effective knowledge reintegration [32]. For example, low-code development facilitated back sourcing in a Scandinavian municipality [33]. Finally, von Bary et al. explore the gaps between research and practice in IS/IT back sourcing [34], and present themes aimed at increasing the practical relevance of future research in the field of IS back sourcing.

To the best of our knowledge, there is neither a comprehensive framework nor a decision support system to help decision makers facing back sourcing issues. In the following, we propose such a framework, which represents a core element of a comprehensive design science research approach, providing decision makers with a set of artifacts that facilitate their back sourcing decision.

3. Research approach

Our research approach is based on the Design Science Research (DSR) paradigm coined by Hevner et al. [12]. DSR consists of the construction and evaluation of artifacts. Although Hevner et al.'s framework offers a global view of the DSR process, it does not provide the operational elements to generate requirements and other deliverables. For this purpose, we complement Hevner et al.'s approach with the

methodologies of Peffers et al. and Kuechler et al. [35,36]. The main steps of our approach are summarized in Table 1.

The first step (Awareness) starts with the initial problem, in this case the need to support the decision maker responsible for outsourced IS/IT processes that she is considering bringing back in-house. It leads to the elicitation of design requirements for a decision support system. Let's remember that design requirements are guidelines that describe the functionalities a product (in our case, a decision support system) must have to meet user needs. The second step (Suggestion) combines these design requirements with systems theory to generate design principles. The latter are value statements that articulate the most important goals a system should fulfil for its users. They are used to inform and structure design decisions. The third step (development) derives a conceptual framework from these design principles. The framework is then translated into a decision tree model, drawing on both academic and professional literature. This is followed by the demonstration step. It draws on organizational theories to operationalize the decision model, from which a prototype is derived. Finally, the evaluation step consists of instantiating the prototype using a case study and gathering expert opinion on this contribution.

3.1. Awareness

The purpose of this phase is to describe the problem at hand, providing the decision maker with a guided process to facilitate the choice between renewing an outsourcing contract (with the same service provider or not) or bringing the process back in-house. Our study of the literature enabled us to elicit four main issues. First, some authors mention the necessity to *obtain a holistic picture* of the back-sourcing process [9,23]. Second, others argue in favor of *structuring knowledge* [37]. Third, the latter supposes that we *build theoretical foundations* [38,39]. Finally, these different elements are necessary to *understand the phenomenon* of back-sourcing [3,17]. Left block of Fig. 1 contains the four issues and a few verbatims from the literature in support of each one.

To obtain the design requirements, we use the requirements ontology adapted to DSR [40]. The ontology distinguishes between functional, non-functional or theory-based requirements, user-based requirements and context-based requirements. It also mentions the requirements adapted from the literature, which do not lend themselves to our situation where we are creating the knowledge base, and those based on features, which are the coarser-grained functional requirements (corresponding here to our 4 issues). In terms of functional requirements, the

aim of our DSS is to organize knowledge and structure it into components such as people, organizational structures and processes (DR1 and DR2). In terms of user requirements, this DSS aims to enable decision makers to share a body of knowledge (DR3) and obtain answers to their questions (DR4). Theory-based requirements must be identified and accumulated over time (DR5 and DR6). Context must also be considered when back-sourcing (DR7). In Table 2 we describe in more detail the seven design requirements shown in the central block of Fig. 1.

Using card sorting, we built a mapping between the four issues and the seven design requirements. The literature review highlighted the need for a holistic approach (LR1 and LR2), the emerging nature of the field (LR3), the need to link to organizational theories (LR4 and LR5) and the complexity of the back-sourcing phenomenon (LR6 and LR7). Thus, issues I1 and I3 lead us to propose *organize back-sourcing knowledge* (DR1) as a requirement. In the same way, the emerging domain of IS/IT back-sourcing generates the need to *facilitate the associated knowledge accumulation* (DR6). In addition, an important requirement is the ability of a back-sourcing framework to *identify relevant theories* and help decision makers support their intuitions with the right theories (DR5), as suggested by LR4 and LR5 (Issue I3). *Structuring the back-sourcing components* is a prerequisite for ensuring the comprehensiveness of the framework (DR2), as required by issue I1. The design of a useful IS/IT back-sourcing framework should consider three additional requirements: *identifying the factors that influence IS/IT back-sourcing* (DR7), as mentioned in LR6 (Issue I4); *identifying the outcomes of back-sourcing decisions* (DR4), derived from issue I4; and *sharing a common understanding of IS/IT back-sourcing* (DR3), implied by both LR1 and LR2.

3.2. Suggestion

Aiming at making suggestions for design principles (DP), we used the design requirements to conceptualize tentative principles that support decision guidance regarding back-sourcing process. These design principles were derived by analyzing the design requirements using systems theory, our core theory, as a guide [41]. This leads us to derive four design principles (DP1 to DP4), summarized in the right-hand block of Fig. 1. Systems theory enables us to understand the system as a whole (DP1, "holistic approach"). It states that a system tends towards a goal (DP2, the "why"). It allows us to describe its structure and activities (DP3, the "what"). Finally, a system evolves in an environment (DP4, the "where"). We then used card sorting to map these four design principles to the design requirements. As an illustration, the combination of *Organize back-sourcing knowledge* (DR1), *Facilitate back-sourcing knowledge accumulation* (DR6) and *Structure the back-sourcing components* (DR2) allows us to derive *Define why this system must exist* (DP2). Similarly, DR3 and DR5 lead to *a holistic approach* (DP1). Once the goal of the system is defined, its components can be structured as induced by DR1 and DR2, leading to DP3: *Define its components and their interrelationships*. Finally, DR4 and DR7 deal with the environment of the system, summarized in DP4: *Define the environment of the system*. The alignment between requirements and systems theory allows us to conclude that the latter is sufficient to structure a conceptual framework.

3.3. Development

The development step aims to derive the decision model. This is done in two phases. The first phase consists of deriving the conceptual framework from the design principles. The second phase uses this framework to build the decision tree.

3.3.1. Concept alignment

Design principles guide the design knowledge on how to build a conceptual framework for a decision support system that meets the design requirements. A framework is an artifact that can be characterized as a system [42]. A system is defined by its dimensions: *goals, structure, activities, environment, and outcomes*. To make this framework

Table 1
Our research approach.

DSR Process Step	Inputs	DSR Activity	DSR Outcomes
Awareness	Initial problem	Literature review	Design requirements
Suggestion	Design Requirements Systems theory Design principles COBIT Metamodel Conceptual framework	Concept mapping Concept alignment	Design principles Conceptual framework
Development	Academic and professional literature Decision tree model Organizational theories Operationalized decision tree	Concept mapping Concept mapping Instantiation	Decision tree model Operationalized decision tree Prototype
Demonstration	Prototype	Illustrative scenario	Evaluation results Performance measures
Evaluation	Case study Framework, model, and prototype Framework, model and prototype	Logical Argument Expert evaluation	Qualitative evaluation

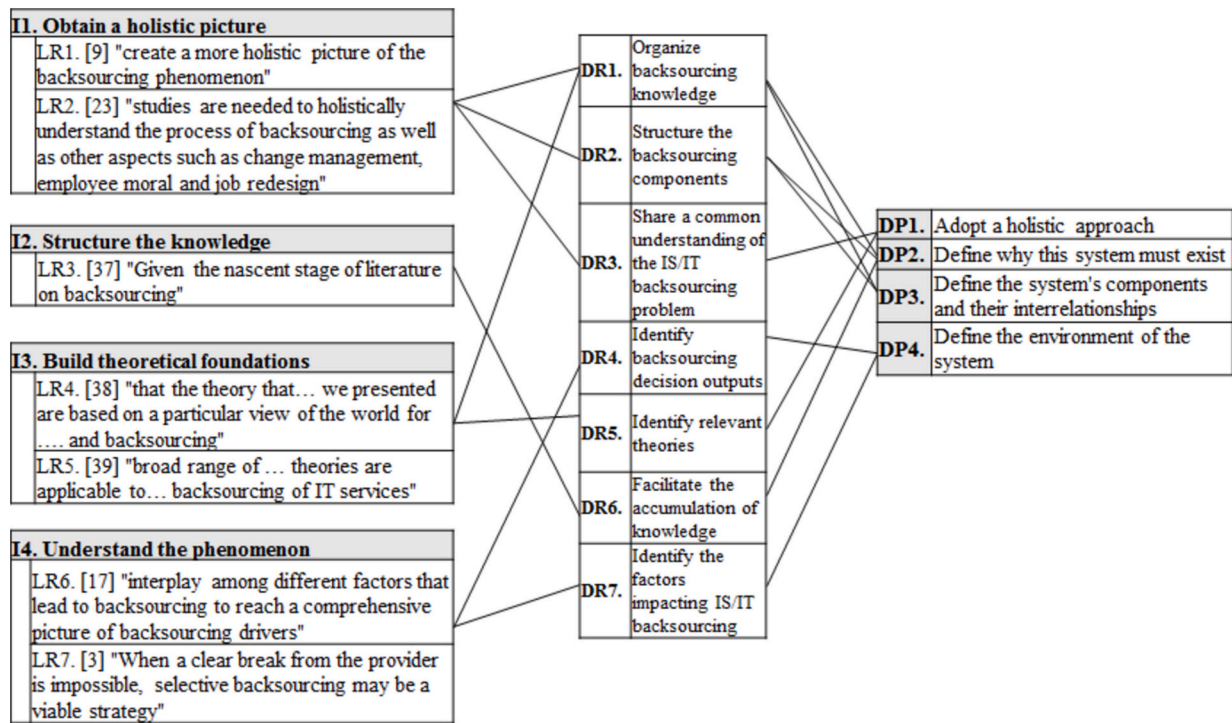


Fig. 1. Overview of the design requirements.

operational, we need to structure each of its dimensions in detail, using available standards or close repositories where appropriate. The IS/IT backourcing decision is a strategic decision related to IS/IT governance. The latter is often structured according to the COBIT 2019 Core Model [43]. Therefore, we performed an alignment process between this core model and the system dimensions to derive the content of our conceptual framework (Fig. 2).

The **Goals** dimension is decomposed into three axes, as suggested by Plugge and Bouwman [44]: the *strategic* axis focuses on the reasons why firms make strategic decisions; the *socio-organizational* axis assumes that the structure of the organization and its environment have an impact on its ability to outsource its IS/IT activities; and the *economic* axis aims to consider all costs (production costs plus transaction costs) when deciding whether to keep services in-house or outsource them. This three-axis analysis has been widely used and validated in outsourcing research, which leads us to recommend and adapt it in backourcing decision making.

The second dimension, **Activities**, identifies the IS/IT processes that are likely to be backsourced. It includes *IT processes*, *IS processes* and *Business processes*. *IT processes* relate mainly to IT infrastructures. *IS processes* are the set of IS applications. *Business Processes* are the set of activities that can achieve the goals described above. They embed and extend the two COBIT components: Information flows and items; Services, infrastructures and applications.

The third dimension is **Structure** since a system is made up of parts and the relationships between them. It corresponds to three COBIT components: *Organizational Structures*, which correspond to the roles played by the various stakeholders; *Policies and Procedures*, which document the rules for the interaction between stakeholders; and *People, Skills and Competences*, which describe the skills required to perform IS/IT activities.

The **Environment** dimension consists of *external factors* that affect the organization. These factors are not included in the core COBIT model. We propose to define this set of factors using PESTLE [45]. A PESTLE analysis (Political, Economic, Social, Technological, Legal and Environmental) examines how these external factors affect an organization. It corresponds to the external context dimension. The

environment also consists of various internal factors that describe the organization and potentially affect the backourcing system. In line with the COBIT core model, we propose to structure these internal factors into three categories: *culture*, *ethics*, and *behavior*.

Finally, the framework includes an evaluation dimension of its activities, called **Outcomes**, described by balanced scorecard components, namely *Financial*, *Customer*, *Internal* and *Learning* [46]. The first component, *Financial*, measures the extent to which the organization works within its budget and makes a profit. The second component, *Customer*, aims to assess whether the organization provides the right services/products to the customer. The third component, *Internal Process*, allows us to assess whether all processes are optimized to deliver the organization's services efficiently. The last component, *Learning and Growth*, ensures that the organization is continually improving. This *Outcomes* dimension allows us to exercise the feedback loop to make the system evolve.

This results in a conceptual framework for backourcing based on systems theory [47]. To the best of our knowledge, this is the first conceptual framework that covers the whole picture. As noted in the requirements statement, this framework can serve multiple purposes. In this paper we illustrate the use of this framework to build a decision tree model and a prototype to help decision makers when faced with a potential backourcing decision.

3.3.2. Concept mapping

Based on this framework, we present below the multicriteria decision model that guides the backourcing decision process (Fig. 3). It provides managers with a more holistic view of the decision process and the ability to evaluate a backourcing decision by combining multiple theories. We have chosen to present the decision model as a hierarchy of criteria [48]. It facilitates the comparison of alternative strategies or scenarios for solving the backourcing problem. It allows criteria to be added, removed, or marked inactive as needed. It allows us to assign weights (w_i in Fig. 3) to the criteria based on the decision makers' perception of the importance of these criteria for a particular case. It provides a dynamic environment to monitor changes that need to be made to achieve the success of the backourcing decision and facilitates

Table 2
The seven design requirements.

DR1: Organize back-sourcing knowledge	There is a body of knowledge about IS/IT back-sourcing, its structure, principles, procedures, and methods. Organization of the knowledge involves the description of information materials related to this body of knowledge that can be used as a basis for the IS/IT back-sourcing decision process
DR2: Structure the back-sourcing components	Companies need to identify the people, organizational structures, and processes involved in back-sourcing projects
DR3: Share a common under-standing of the IS/IT back-sourcing problem	This requirement relates to the design of a controlled back-sourcing vocabulary that supports the exchange of information between stakeholders involved in the decision and its implementation, and a mutual understanding of the concepts and methodologies used in IS/IT back-sourcing frameworks
DR4: Identify the results of back-sourcing decisions	The conceptual framework structures back-sourcing and improves accountability by reporting on planned and actual performance. This requirement ensures that the organization's IS/IT back-sourcing decision aligns with the corporate policy agenda
DR5: Identify relevant theories	There are many theories that are relevant to the back-sourcing decision process. These include agency theory, resource dependence theory, institutional theory, and systems theory. This requirement aims to identify the most relevant or appropriate theories for IS/IT back-sourcing
DR6: Facilitate the accumulation of knowledge	We aim to understand how this body of knowledge can be used to develop our conceptual framework and for theorizing. Decision makers will be able to define a new and relevant strategy by capitalizing on the knowledge gained, particularly the theories and practices of IS/IT back-sourcing
DR7: Identify factors that influence IS/IT back-sourcing	We aim to identify the factors influencing IS/IT back-sourcing. These include regulations, culture, information architecture, and compliance systems. This will help us understand the current outsourcing situation and identify areas for change.

sensitivity analysis. In fact, the nodes of the hierarchical decision model can have variable weights and their number is not limited.

The purpose of the decision tree is to organize the evaluation of the activity likely to be reinternalized. It makes it possible to prioritize the evaluation criteria. Each activity therefore requires the use of this hierarchy. Based on the framework (Fig. 2), these activities may include IT processes that manage hardware, software, network technologies, and data; IS processes and the associated people involved in IS creation, maintenance, and/or management; and business processes and their associated IT technologies (ACTIVITIES, first level of the hierarchy shown in Fig. 3).

The systems theory vision proposed in the framework allows the criteria to be organized by component. For each activity, the most structuring level is the goal dimension, as mentioned by many authors, including Plugge and Bouwman [44]. Thus, GOALS, the second level of the hierarchy, allows us to analyze each activity in terms of the three different types of goals associated with the system. The first dimension, *strategic* analysis, refers to the identification of activities in need of improvement, including back-sourcing. It is an internal strategic analysis that allows companies to identify the set of resources that can be used to improve the company's image and performance. This includes assessing the potential for back-sourcing some activities. In addition, external strategic analysis allows companies to better understand the external factors that may hinder their growth. The second dimension, *socio-organizational* analysis, allows companies to assess where the organization currently stands and provides information on the most promising

way to implement new growth strategies. It covers all aspects of organizational change, including structure, people and policy. The third dimension, *economic* analysis, aims to assess costs and benefits. It ranks alternative scenarios on the basis of economic viability and optimizes the allocation of resources.

The criteria that appear as the fifth level of the tree (CRITERIA) are the result of identifying all the expertise available in the literature to support back-sourcing decisions. Some criteria will be more important to some stakeholders than to others. Some may be irrelevant in some contexts. We therefore present a comprehensive set of generic criteria relevant to back-sourcing evaluation. It can be proposed as a repository from which stakeholders can identify the subset of criteria appropriate to the back-sourcing problem. The analysis of this set of criteria has enabled us to inductively relate them to the three categories of goals. The first eight criteria fall into the strategic goal category. The last four are of an economic nature. The others relate to organizational structure or the environment. Through a card-sorting process, the *Structure* and *Environment* components of the system and their sub-dimensions emerged, allowing the relative weights of these to be associated during the evaluation. We can thus see that this process, combining a top-down and bottom-up approach to the construction of the tree, allowed us to locate all the components of the system, except for the *Outcomes* dimension, which will be mobilized during the operationalization of the tree (Section 3.4).

As far as the socio-organizational analysis is concerned, the conceptual framework allowed us to enrich the hierarchy with two additional intermediate levels to better organize the components. The first intermediate level is based on the *Structure and Environment* components of the conceptual framework (STRUCTURE & ENVIRONMENT, third level of the hierarchy in Fig. 3). The Structure branch is broken down into three components: *Organizational Structures*; *People, Skills and Competencies*; and *Policies and Procedures*. Finally, the Environment branch is subdivided into *External Environment* and *Internal Environment* (STRUCTURE & ENVIRONMENT SUBDIMENSIONS, fourth level of the hierarchy).

All the criteria in the tree relate to problems or opportunities. Each criterion can be analyzed using one or more theories, as explained in Table 2. Problems that cannot be solved with the current outsourcer and new opportunities that may change the relevance of the existing outsourcing strategy are the two main types of reasons that are used to question a potential back-sourcing [3]. For example, regarding the *organizational structure* sub-dimension, 'management change' represents a problem and/or opportunity to be considered when a decision maker is faced with the question of back-sourcing. Another example, in relation to the *strategic* goal, is 'know-how mismatch', which considers whether there is a know-how problem. Thus, the know-how mismatch criterion invites the decision maker to assess whether such a problem has been identified. This occurs when the outsourcer becomes more proficient in the outsourced activity than the company itself, according to the resource dependence theory. As far as the *economic* goal is concerned, 'Excessive cost' assesses the existence of problem that stems from the cost of the existing outsourcing solution.

Finally, the last level of the tree represents the three SCENARIOS (Renegotiate, Change Vendor and Backsource) that will be evaluated against the criteria. The decision tree organizes the criteria that help the decision maker analyze the outsourced process. To facilitate its implementation, each criterion must be made operational. This is one of the objectives of the demonstration step.

3.4. Demonstration

This step consists of two phases. In the first phase, each criterion of the tree is enriched with one or more theories to facilitate its evaluation by the decision maker. In the second phase, a prototype is designed to automate the global evaluation of an outsourced process.

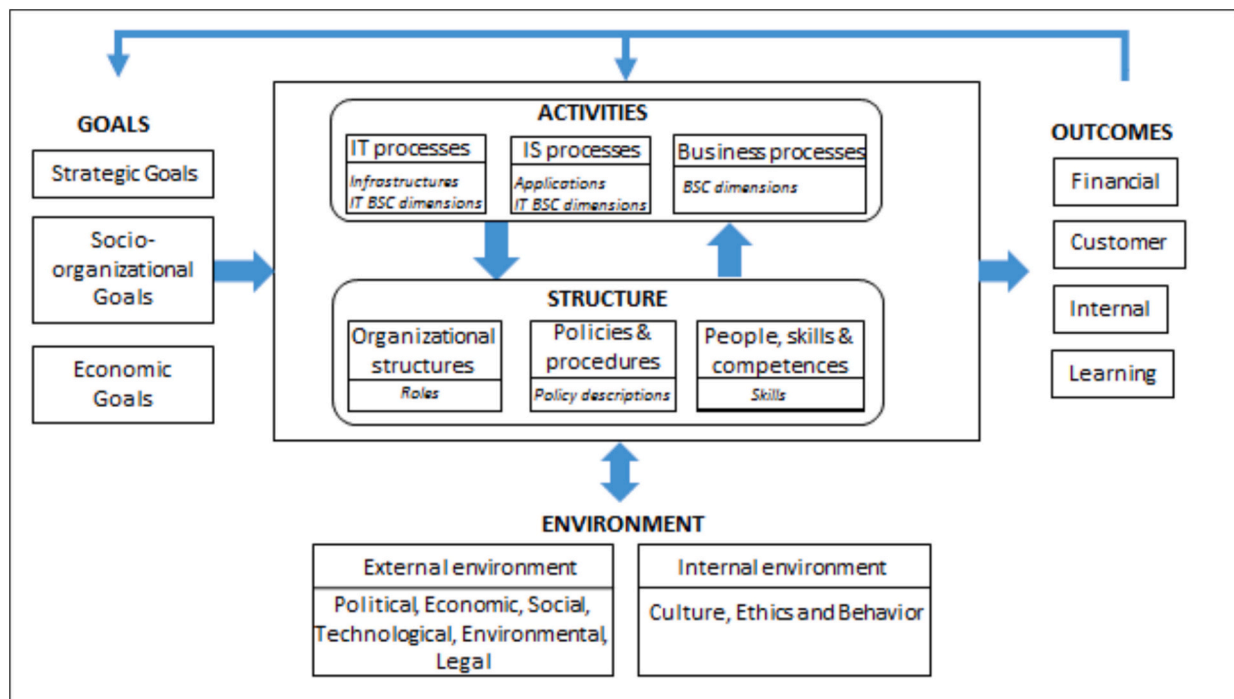


Fig. 2. The conceptual framework.

3.4.1. Concept mapping

Due to the nature of the tree structure, a decision maker needs to evaluate the criteria to be applied to his or her situation. Using one or more theories helps to achieve this. In the literature (both professional and academic), we looked for recommendations to backsource, or to change supplier, or to maintain the status quo, with the arguments provided. Sometimes these arguments (especially in the academic literature) were accompanied by a theory. In other cases, we had to search for the most relevant theory or theories to support the recommendation.

By associating the relevant theory with each criterion, we provide decision makers with a tool to inform them of what to do. For example, a decision maker may use social exchange theory to evaluate the "poor service quality" criterion, because as service quality declines, one option for the customer is to exit the relationship and backsource. Table 3 provides a brief description of some of the theories that can be applied to the issue of back sourcing. The criteria are listed in the first column. The criteria highlighted in orange represent problems that could lead to back sourcing, while those highlighted in green represent opportunities. An explanation of their meaning in the context of back sourcing is suggested in column 3. The main theory associated with each criterion is listed in column 4. Column 2 describes the recommendations resulting from the application of the theory and lists the possible scenarios between: back sourcing the activity, changing the supplier or renegotiating the contract. Table A (in the appendix) provides a complementary list of criteria.

As can be seen in Table 3, many theories have been mobilized to understand and explain IS/IT sourcing decisions. Plugge and Bouwman, who studied the outsourcing problem, suggested the following theories: resource view of the firm, transaction cost theory, competency theory and organizational theory [44]. Following Shook et al., we extend this list to include theories such as institutional theory, resource dependence theory, systems theory, agency theory and innovation theory [49]. These theories, which apply to the problem of sourcing in general, also apply to the problem of back sourcing. By multiplying the number of theories, we increase the possibility of analyzing the back sourcing problem from different perspectives. These theories support the decision

maker's cognitive process when evaluating the criterion. Below we present some of the theories relevant to each dimension (second level of the hierarchy in Fig. 3), namely *strategic*, *socio-organizational* and *economic*. Space does not permit us to present them all (they are summarized in Table A of the Appendix).

Strategic analysis can be done using the theory of core competencies [31]. The latter explains that the core competencies of the company must not be outsourced. The importance of an activity may change over time. A competence that becomes or returns to the core business must be brought back in-house. The second type of strategic analysis, environmental pressure analysis, refers to institutional theory. The latter explains how an organization's environment institutionalizes and legitimizes strategies through regulative, normative, and cognitive mechanisms [50]. This theory predicts that institutional pressures can lead the organization to adopt back sourcing strategies that are in line with its environment. The third type of strategic analysis is related to dependency analysis. Resource dependence theory (RDT) allows organizations to analyze how they control their IS/IT resources in order to improve organizational performance. RDT posits that organizations must limit external dependencies. "The more critical the resource is to the organization, the more serious the dependency" [51]. Back sourcing is a solution that limits or even eliminates this dependency. Finally, the fourth type of strategic analysis is competence analysis using competence theory. The latter characterizes a firm as an open system of tangible and intangible (e.g., competencies) assets, stocks and flows [52]. IS literature suggests that firms' competencies are positively related to its performance [53]. Having internal competencies facilitates a back sourcing decision that leads to an improvement in firm performance.

Socio-organizational analysis includes systems analysis, cultural context analysis, and organizational structure analysis, which can be conducted using related theories. Systems analysis can be conducted using systems theory. The latter views the organization as a system of interrelated parts that interact in the production of products and services [54]. From a systems perspective, it is assumed that if one part of the system is back sourced, this should lead to the back sourcing of other interconnected parts of the same system. Cultural context analysis is

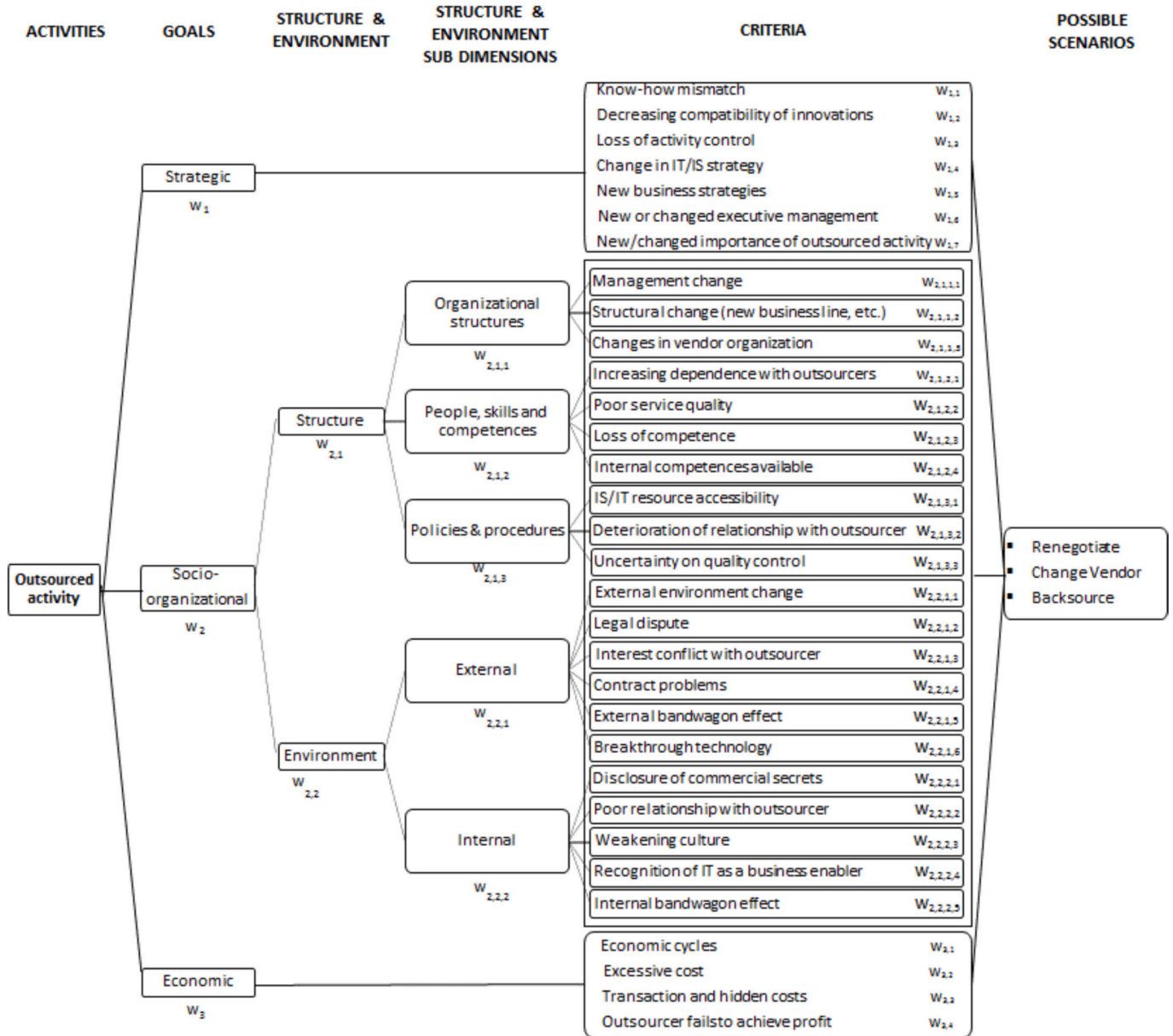


Fig. 3. The decision tree.

related to the innovation relationship between outsourcers and firms. The cultural context of an organization influences its ability to innovate. The theory of innovation posits that if the outsourcer does not provide the means to introduce the innovation desired by the client company, then recourse to back-sourcing becomes prudent [55]. Organizational structure analysis refers to agency theory. According to this theory, an agency is a relationship between two parties in which the agent represents the principal in some transactions [56]. In an outsourcing relationship, the agent (vendor) and principal (customer) must have common goals and the same degree of willingness.

Finally, *economic* analysis can be divided into cost analysis and profit analysis. Cost analysis refers to transaction cost theory (TCT). The latter is the predominant theory used to examine firms' sourcing decisions from a make versus buy perspective [57]. TCT tenets imply that sourcing decisions involve a comparison of the production costs associated with producing a process/product internally (hierarchy) with the transaction costs associated with purchasing a process/product from an external source (market). Profit analysis is related to profit maximization and competitive theory [58]. The latter posits that the organization's

primary objective is to maximize long-term profits and develop a sustainable competitive advantage. This is the case if the outsourcer is not making a profit.

The construction of this decision tree allows us to look at back-sourcing issues from different perspectives. We provide managers with multiple insights into the back-sourcing problem, facilitating their decision-making process. Since multiple factors from both the internal and external environment contribute to the back-sourcing decision (cost savings, dissatisfaction with outsourced services, changes in the external market, etc.), the need for a holistic perspective is useful.

3.4.2. Instantiation

In the following, we present the prototype that implements the decision model described above. Our approach is operationalized using the Super Decisions software [59]. The latter implements the Analytic Hierarchy Process (AHP), which allows us to combine judgments and data to rank alternative scenarios and predict outcomes. This tool can be used to efficiently compare back-sourcing alternatives. It can also be used to provide feedback to stakeholders in the decision process. Super

Table 3
Possible scenarios and relevant theories.

Criterion	Possible scenarios	Explanation	Relevant theory
Know-how mismatch	• Change Vendor • Backsource	The outsourcer knows more about the IS/IT process than the company. Over-dependence on the IT supplier is considered a major outsourcing risk	Resource dependence theory
External business changes	• Renegotiate • Change Vendor • Backsource	Mechanisms in place that monitor the environment for emerging technologies that may provide strategic advantage are difficult to outsource	Institutional theory
Decreasing compatibility of innovations	• Change Vendor • Backsource	Outsourcing can hamper IS/IT innovation	Theory of innovation
Loss of activity control	• Backsource	To avoid a loss of control, companies should avoid outsourcing the IS/IT elements critical to their success	Strategic contingencies theory
Changes in IS/IT strategy	• Renegotiate • Change Vendor • Backsource	Mechanisms in place that monitor the environment for emerging technologies that may provide strategic advantage are difficult to outsource	Institutional theory
New or changed executive management	• Renegotiate • Change Vendor • Backsource	Arrival of a new executive member	Power and politics theory
New business strategies	• Backsource	Organizations must constantly adapt to their competitive environment in order to survive	Survival-based theory
New/changed importance of outsourced activities	• Backsource	A core activity should not be outsourced	Theory of core competencies

Decisions allows us to present the set of criteria described above through a hierarchical structure. It allows criteria to be grouped within similar dimensions. It also allows criteria to be added and removed as needed without distorting the decision process. The resulting hierarchy makes it possible to evaluate the back sourcing problem at any level. For example, management can view the evaluation from a *strategic* perspective or from the totality of the three dimensions (strategic, socio-organizational, economic). Criteria can be measured on a scale. Stakeholders can weigh the importance of each criterion and rate the back sourcing problem in terms of the criteria at lower levels of the hierarchy. These ratings can be aggregated throughout the hierarchy, and each node is assigned a rating score. Fig. 4 shows the hierarchy of our decision model using Super Decisions software. The latter is a software package developed for the analysis of complex decisions based on the AHP methodology [48].

The root corresponds to the goal, in this case the decision to backsource an activity. The second level is the criteria hierarchy with its three branches (strategic, socio-organizational, and economic). The other levels implement the hierarchy shown in Fig. 3. Thanks to this hierarchy, the three alternatives (renegotiate the contract, change vendor, backsource the business) are evaluated. The process must be applied to each outsourced activity.

In this section, we have successively described the first four steps of

our approach, including the four artifacts we contributed. A first evaluation process was performed. It is described below.

4. Evaluation

Peffer et al. describe a number of evaluation methods for DSR approaches according to the proposed artifacts (algorithm, method, model, etc.) [60]. On this basis, we selected three complementary methods for our context. First, we illustrated the approach by applying it retrospectively to a case study published on IT back sourcing [27]. Second, we developed a logical argument comparing our approach with twelve recommendations from the literature for an IT outsourcing decision support approach [11]. Finally, we presented our approach to a panel of experts to get their point of view. Thus, we evaluate the efficacy, effectiveness, and utility of our approach [42].

4.1. Illustrative scenario

We performed a first validation of our approach based on the Teleco case described and analyzed by Law [27]. After a period of outsourcing several activities, the company Teleco was led to backsource some of these activities, first as an emergency measure and then in a more organized way. Teleco is a company in the telecommunications sector that had outsourced many IS/IT activities in 1996 for a period of 10 years. In 2006, a crisis led the company to question this choice. We analyze two IS activities, Alpha and Beta, and one IT exploitation activity, Omega, that were outsourced. The approach is applied successively to each activity.

Alpha is a key information system for Teleco, as it contains interfaces, mainly with media (TV) partner systems. In the media industry, competitive and technological pressures forced Teleco to rapidly evolve the Alpha system. The outsourcer was not up to the task in terms of know-how and was not able to provide the skills needed to launch the new projects required by the context. There were two options: change vendor or set up a back sourcing process. In addition, Teleco still had in-house skills that would make back sourcing more efficient than outsourcing to another provider. According to institutional theory, the solution has to fit the environment: the media partners require Teleco to have a more scalable information system, which the current outsourcer cannot support. It seems that other potential outsourcers could not be operational quickly. Therefore, this alternative cannot be implemented quickly. As a result, Teleco had to backsource using internal capabilities. Another way of reading the situation is through competence theory. The “know-how mismatch” of the current outsourcer can be interpreted with the help of resource dependence theory, which argues that over-dependence on the IT supplier is considered a major outsourcing risk. In the case of Teleco, there is an opportunity for internal competence, which is why back sourcing is recommended.

The Beta activity is not satisfactory in terms of outsourcing because the control of the outsourcer does not seem to be satisfactory. Moreover, the decision to backsource Alpha had already been made, so the back sourcing of Beta seemed logical since a bandwagon effect had been identified. According to the human resource theory, the strategy development of organizations depends heavily on human factors. Thus, only the maintenance of Beta was back sourced to better control the vendor. Let's note that we do not have all the information about the difficulties encountered in managing the contract with the outsourcer. The analysis may lead to the identification of non-aligned interests between Teleco and the outsourcer (agency theory) or a problem of hidden costs leading to an imbalance between production costs and transaction costs as used in transaction cost theory. However, Law highlights the difficulties that this partial back sourcing will encounter due to the lack of internal skills (for reasons of space, this second cycle of back sourcing analysis will not be developed here).

Omega is Teleco's IT exploitation activity. The cost analysis showed that the outsourcing contract was expensive. In addition, there were

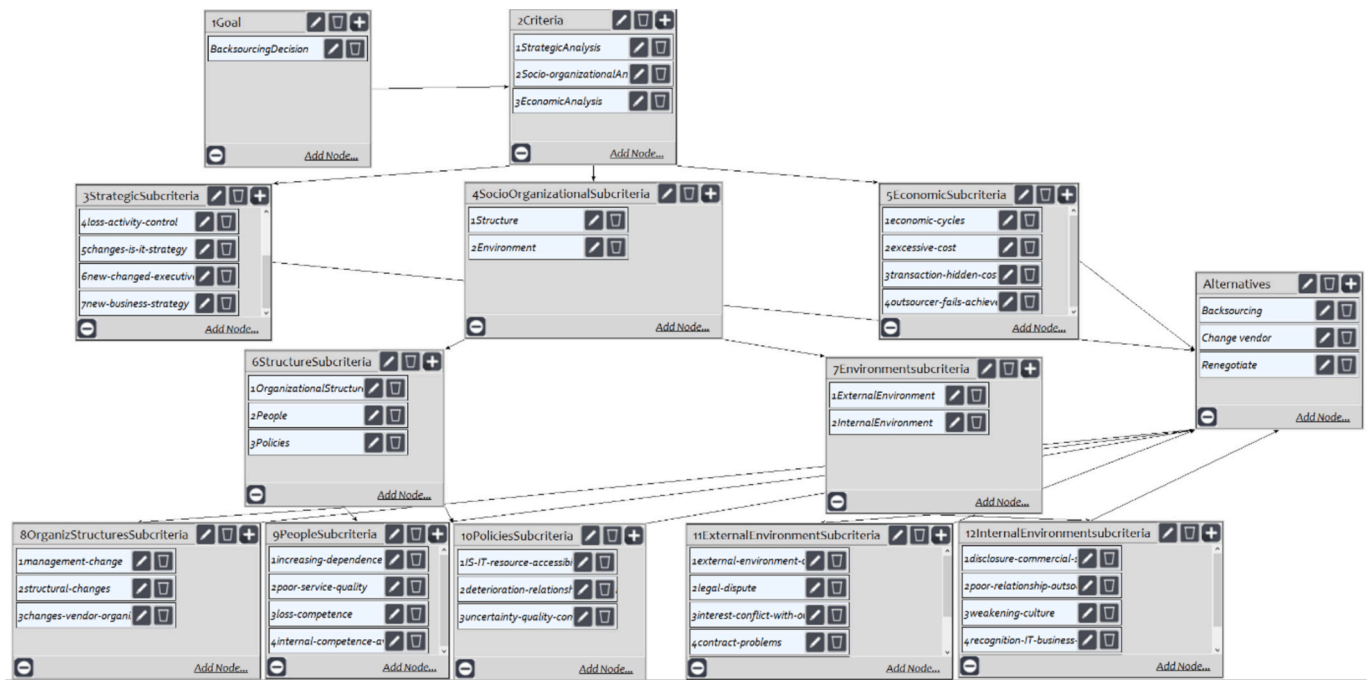


Fig. 4. A screenshot of the AHP model using Super Decisions software.

difficulties in managing the contract. This led Teleco to review the reversibility and cost of the contract. A renegotiation followed, resulting in selective backsourcing. This led to the creation of an internal structure for this selective backsourcing. Cost theory suggests that a backsourcing solution should be preferred when the transaction costs (outsourcing contract) are higher than the production costs (insourcing costs). The bandwagon effect of other activities being backsourced is an opportunity to restaff IT teams around a consistent set of activities (human resource-based theory). By viewing IS as a system with multiple interdependent components (systems theory), the spillover effect helps to globalize the backsourcing decision.

The informed argument developed above helped to illustrate ex post how Teleco's decisions can be illuminated by several theories, including human resource-based theory, competence theory, institutional theory, resource dependence theory, agency theory, systems theory, and transaction cost theory. Space does not permit us to illustrate all the theories mentioned in our approach. Moreover, as the Teleco example has shown, several theories may be relevant in the same situation to recommend or discourage backsourcing. There are also situations where theories associated with different dimensions contradict each other regarding the relevance of backsourcing. In this case, weighting can play a role in prioritizing the recommendations.

Applying our approach to the Teleco case yields the results shown in Table 4. The first column shows the criteria involved in the decision to backsourcing an IS/IT activity. The second column shows the weight of each criterion according to its hierarchical level. Thus, the strategic, socio-organizational and economic criteria have a weight of 0.4, 0.3 and 0.3 respectively. Obviously, the sum of the weights at each level equals 1. The weights are determined by mutual agreement between the decision maker and the expert, based on past experience. In some cases, we could use a database of weights chosen in similar situations. We do this for all levels of the hierarchy in Fig. 3. The third column corresponds to the valuation of Teleco's Alpha business, which is likely to be the subject of a backsourcing scenario. For the Alpha business, the relevant criteria are: "Loss of activity", "Poor service quality", "Internal competences available", "Deterioration of relationship", "External environment change", "Contract problems", "Internal bandwagon effect", and "Transaction and hidden costs". These cells are marked "yes". The next

three columns indicate the type of scenario to be selected (renegotiation of outsourcing contract, and/or change of provider, and/or backsourcing) for the Alpha activity. At the intersection of the line corresponding to the selected criterion and the type of scenario is the value corresponding to the choice (0 or 1). A value of 0 corresponds to the rejection of this scenario. Conversely, a value of 1 corresponds to the selection of this scenario. For example, for the criterion "Loss of Activity Control", the only scenario selected is "Backsource". We apply this procedure only to the terminal nodes of the hierarchy. The calculation of values for intermediate nodes follows the AHP rules, which consist in calculating the weighted average (value*weight) at each level of the hierarchy. For example, for the strategic level, the respective weights of the criteria are: [0.2;0.1;0.1;0.1;0.2;0.1;0.1;0.1]. Only the "Loss of Activity Control" criterion has a value of 1, so the value of the "Strategic" node is $1 \times 0.1 = 0.1$; continuing in this way for the entire tree structure, the evaluation value for the Alpha activity is 0 for "Renegotiate", 0.039 for "Change Vendor" and 0.196 for "Backsource". Doing the same for the other two activities (Beta and Omega), we get the results shown in Table 4. The last row of the table shows the results. Our recommendation is to consider the Backsourcing solution first for each activity, since this scenario achieves the highest score for all three activities. This score is relative, and its legitimacy is limited to a comparison with the other two possible scenarios. A change of vendor scenario is not totally out of the question for all three activities. Finally, a renegotiation of the outsourcing contract is only conceivable for the Beta business. In addition to the score assigned by the tool, the list of criteria that apply to the activity provides a set of arguments on which the decision maker can base his or her recommendation. Our decision support system enables decision makers to strengthen their arguments. It provides a basis for discussion among stakeholders, leading to a solution that is acceptable to the majority. In addition, our prototype can be used to simulate different scenarios. It can also be used to measure the consequences of alternative criteria and their respective weights. Furthermore, by applying our approach and tool to several backsourcing projects, we can gradually build up a database that can be used as a basis for criteria and weights.

This validation effort through an illustrative case study is the first step in a more ambitious process to fully validate the approach proposed

Table 4
Application of the AHP approach to the case study.

Criterion	Weight	Alpha activity				Beta activity				Omega activity			
		Review	Rene-gotiate	Change Vendor	Back-source	Review	Rene-gotiate	Change Vendor	Back-source	Review	Rene-gotiate	Change Vendor	Back-source
Strategic	0.4		0	0	0.1		0.2	0.2	0.2		0	0.2	0.3
Know-how mismatch	0.2									yes	0	1	1
Ext. bus. Changes	0.1												
Decreasing compatibility of innovations	0.1												
Loss of activity control	0.1	yes	0	0	1					yes	0	0	1
Changes in IS/IT strategy	0.2					yes	1	1	1				
New or changed exec. Mgt	0.1												
New bus. Strat.	0.1												
New/changed importance of activities	0.1												
Socio-organizational Structure	0.3		0	0.132	0.22		0.072	0.084	0.252		0	0.18	0.228
Organizational structures	0.6		0	0.14	0.22		0.12	0.14	0.38		0	0.14	0.22
Management change	0.4												
Struct. changes	0.4					yes	1	0	1				
Changes vendor organization	0.2									yes	0	1	1
People	0.4		0	0.2	0.4		0	0.2	0.2		0	0.2	0.4
Increasing dependence	0.2												
Poor service quality	0.2	yes	0	1	1	yes	0	1	1	yes	0	1	1
Loss of competence	0.4												
Internal competences available	0.2	yes	0	0	1					yes	0	0	1
Policies	0.3		0	0.2	0.2		0	0.2	0.6		0	0	0
IS/IT resource accessibility	0.4												
Deterioration of relationship	0.2	yes	0	1	1	yes	0	1	1				
Uncertainty on quality control	0.4					yes	0	0	1				
Environment	0.4		0	0.12	0.22		0	0	0.06		0	0.24	0.24
External	0.4		0	0.3	0.4		0	0	0		0	0.3	0.3
External environ. Change	0.1	yes	0	0	1								
Legal dispute	0.1												
Interest conflict with outsourcers	0.1												
Contract problems	0.3	yes	0	1	1					yes	0	1	1
Ext. bandwagon effect	0.1												
Breakthrough technology	0.3												
Internal	0.6		0	0	0.1		0	0	0.1		0	0.2	0.2
Disclosure of commercial secrets	0.2												
Poor relationship with outsourcer	0.2									yes	0	1	1
Weakening culture	0.2												
Recognition of IT as business enabler	0.3												
Int. bandwagon effect	0.1	yes	0	0	1	yes	0	0	1				
Economic	0.3		0	0	0.3		0	0	0.3		0	0.2	0.5
Economic cycles	0.2												
Excessive cost	0.3					yes	0	0	1	yes	0	0	1
Transaction and hidden costs	0.3	yes	0	0	1								
Outsourcer fails to achieve profit	0.2									yes	0	1	1
Evaluation			0	0.039	0.196		0.101	0.105	0.245		0	0.194	0.338

here.

4.2. Logical argument

In addition, we performed a comparison of our approach with the recommendations proposed by Rajaeian et al. [11]. As it can be seen in Table 5, our approach meets nine of their twelve recommendations. In particular, Rajaeian et al. point out the need to adopt a rigorous research approach, with a state of the art of both academic approaches and professional practices, an elicitation of the requirements, a validation of the approach, and a highlighting of the limitations. In addition, the DSR approach is recommended. The consideration of the organizational context is advocated. Let's mention that our approach, as it stands, does not implement a group decision support system, nor does it provide a full validation of the approach in a real setting. We have highlighted the

Table 5
Validation through recommendations.

Rajaeian et al. recommendations [11]	Implementation of the recommendations
Artifact should be capable of group decision support and capable of supporting C-level managers	The current approach does not consider group decision support. However, future research will address this aspect, knowing that AHP is adaptable to group decision support
Organizational context and characteristics should be considered in the artifact design and development process and should be explicitly presented	Our approach integrates the organizational context and its characteristics into its model
DSS artifacts should be grounded in organizational decision-making research	Our framework is based on systems theory and our decision tree model implements criteria based on organizational theories
Decision variables incorporated in the designed artifact should be derived from IT outsourcing (ITO) descriptive literature and grounded in IT outsourcing reference theories	Our decision variables (i.e., all criteria in the decision tree model) are derived from the outsourcing and back-sourcing literature and are grounded in organizational theory (Tables 2 and A in the Appendix)
ITO decision support frameworks, instruments, models, constructs, methods, and instantiations available in the literature should be reviewed and critically assessed	The literature review in Section 2 discusses and evaluates previous frameworks and other artifacts. Based on this assessment, we derived design requirements that guide our approach
Practice-oriented research methodologies that consider both rigor and relevance such as design science research, action research and case study should be used for development of ITO DSS	We embraced design science research for its rigor and relevance
ITO decision support artifact should be validated	Our artifacts are validated by the means of an illustrative scenario (based on a case study through an ex-post analysis), enriched by the expert evaluation described in Section 4.3
ITO decision support artifact should be verified	Future research will address this aspect
Requirements for implementation of the artifact including usability, readiness for use should be addressed	The relevance of the approach is obtained by eliciting 7 design requirements. The utility has been partially validated. Usefulness is partially addressed in Section 4.3
Assumptions and limitations of the artifact, its appropriate use, and the logic of decision model should be presented	The paper describes the assumptions and limitations of the approach, as well as its use and basic decision model logic
Naturalistic (field setting) evaluation should be used	We plan to conduct a field setting evaluation in future research to better validate the approach
Appropriateness of the decision analysis method(s) selected for DSS should be justified	The decision analysis methodology is derived from systems theory, which was used to develop the conceptual framework for the approach

implemented recommendations in gray.

4.3. Expert evaluation

To test the utility of our approach, we interviewed three practitioners. Our evaluation efforts focused on the approach and the decision model. Due to space limitations, we report only the key elements from the three complementary interviews below.

Our first interviewee oversees third-party application maintenance contracts for a subsidiary of a large European automotive company, having previously worked in the defense and nuclear industries for a total of eight years of outsourcing experience. In her current position, she has not personally had the opportunity to make the decision to reinternalize, which in the cases she knows of is dictated top-down by successive subsidiary managers. The latter are appointed for three years and are not really in favor of the long term. Her testimony allowed us to validate the strategic and economic branches of the decision tree. She sees the model as an invaluable tool for broadening reflection and discussion within committees when such issues arise. In particular, the loss of internal competencies during outsourcing seems to her to be a real problem, which alone or in combination with others could provide a strong justification for re-internalizing IS processes.

Our second interviewee has thirty years of experience in IT services, both as a consultant to large companies and in internal responsibility in many large structures. During the interview, he mentioned a large international NGO in the health sector, a large bank, a telephone operator, and a large industrial company. In his argument, he emphasizes the importance of the reversibility plan and the definition of the scope of the activities at hand. Examining our approach and the decision model, he found interesting the hierarchical organization of the information, its help in assessing the situation and exploring the axes of analysis, the weighting, which makes it possible to adjust and refine the relative importance of both the axes and the criteria. In particular, he confirmed the existence of the PESTLE criteria that lead certain companies to review contracts with actors located in risky countries. From his point of view, the control of contracts is an essential element that resonates in our tree structure at the level of the external environment. He also insisted on the need to take into account the cost of bringing the activity back in-house, which is often a hidden cost in the decision-making process (transaction and hidden costs criterion in the economic sector). Our interviewee particularly emphasized the importance of regulation and respect for international law, which constrain companies in their ability to implement outsourcing of certain activities. These aspects are present in the external environment branch of the tree (legal disputes and contractual problems). He suggested that in a future version of the model, we should better emphasize aspects related to security, which, in his opinion, should polarize decisions more.

The third interviewee has extensive experience in IS outsourcing and back-sourcing, having worked for several consultancies. Given his experience, he found the approach highly relevant. He found in the decision model the criteria he used to make a decision in the situations he experienced, which led him to recommend re-internalization in the first case, to change the provider in the second, and to maintain the status quo in the third.

This comparison of our approach with experts in the field confirms its utility. Depending on the interlocutor, the parts of the tree on which he focuses his remarks may differ, due to his own experience, profile, and context. The combination of points of view reassures us in this multi-criteria approach, which is conducive to the exchange of points of view.

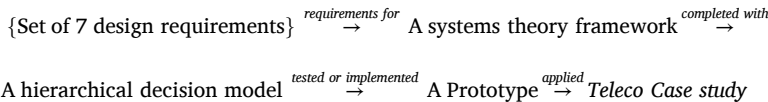
5. Discussion, conclusion, and further research

The work presented in this paper relates to the design, development and validation of a model to facilitate the IS/IT back-sourcing decision process. Following a DSR approach, this paper describes a set of artifacts

used as pillars for an approach to assist decision makers facing a potential decision to backsource their outsourced IS/IT activities. A thorough literature review informed the different artifacts: a set of requirements to be met by a conceptual framework based on systems theory, a decision tree based on the Analytic Hierarchy Process, and a prototype that implements the decision process and delivers a set of weighted recommendations regarding back sourcing issues. In order to make the requirements and the framework IT-ready, we implemented our approach in a prototype using Super Decisions software. A validation based on an illustrative scenario, a logical argument and an expert evaluation is presented. In addition, we explained how our approach meets the main recommendations found in the literature.

Theoretically, this research fills the gap in the literature by adding to the growing body of knowledge on the phenomenon of back sourcing. Practically, our approach contributes to providing practitioners with a decision tree to facilitate their back sourcing decision process. In terms of managerial contribution, we provide managers with a structured analysis covering the three strategic, organizational, and economic dimensions. One practical contribution of our study, based on four artifacts including a multi-criteria decision tree, is to broaden the decision makers' reasoning by confronting them with the different facets of their context. As a result, the decision is more robust and can be based on successive simulations of different scenarios. Our multi-criteria decision support approach is also conducive to sensitivity analysis.

Our research project began with an in-depth literature review with the dual objectives of: 1) confirm the need for such a decision support system for back sourcing IS/IT, and 2) collect elements to build the knowledge base of this system. The state of the art has allowed us to present in this paper an original overview of back sourcing research, its foundations, content and logic. It identifies the papers that play a predominant role in understanding the field. This provides a sound basis for doctoral students and novice researchers who are considering getting involved in this field of research. In our case, it has motivated us to complement the existing research, which is often based on case studies and qualitative approaches. In addition to synthesizing this literature, we extracted from it the rules that dictate the choice of back sourcing and the context in which they can be applied.



Taking advantage of the knowledge thus available, and with the aim of developing a decision support system available to practitioners, we adopted a DSR-type approach leading to a prototype of this future system. Specifically, we produced four artifacts that constitute a decision support approach to IS/IT back sourcing. One of the advantages of our approach is its holistic view of the problem. It allows us to consider all aspects of the back sourcing decision. The first artifact of this approach draws on the literature to extract the design requirements for the future system. The second artifact is the conceptual framework based on the systems theory approach. This framework plays an essential role in the development of the decision model, which is the subject of the third artifact. This hierarchical multicriteria model guides the back sourcing decision process, providing managers with a holistic view of the decision process and the ability to evaluate a back sourcing decision by combining multiple theories. Finally, the fourth artifact is a prototype that implements the decision model. Our approach is implemented using the Super Decisions software. The latter implements the AHP, which allows us to rank alternative scenarios.

Our approach integrates and extends the frameworks mentioned in

Table 6
Comparison of our approach with previous ones.

Reference	Previous approaches	Our approach
Akoka et al. [2]	framework for understanding the back sourcing phenomenon	framework for structuring the back sourcing decision
Felser et al. [5]	approach based on a knowledge-based view of the firm, a foundation for research framework as a tool to examine whether back sourcing is beneficial based on costs and opportunities for re-internalization	a systems-theory framework to inform back sourcing decisions
Thakur-Wernz [19]	framework as a tool to examine whether back sourcing is beneficial based on costs and opportunities for re-internalization	framework as a tool based on more criteria
Slynstag [22]	the framework is a 4-step back sourcing process (including a scoping step) that is not specific to IT/IS processes	our approach focuses on the scoping step to decide which IS/IT processes should be back sourced
McLaughlin and Peppard [25]	an end-to-end outsourcing decision model with 3 options (renew, backsource, change vendor)	our multicriteria decision approach includes these 3 options as 3 possible scenarios to compare
Solli-Sæther et al. [26]	their stages-of-growth model is proposed as a theory to be explored and empirically validated, or as a maturity model for practitioners. It includes three axes of analysis	we include the 3 axes of analysis in our set of criteria to help the back sourcing decision maker

Section 2. We highlight its added value in Table 6.

To our knowledge, this is the first time that a sound approach based on DSR and systems theory has been used as a tool for the back sourcing decision-making process. A set of theories is used to enhance stakeholders' reasoning in the decision-making process. As suggested by Akoka et al. in order to facilitate the dynamic summary of our contribution, we present below the path of knowledge types of this paper [61]. More precisely, we have adapted the concept of the path of knowledge types with five nodes denoting the four artifacts and the Teleco case study:

This path of knowledge type allows us to show the dynamics linking our four artifacts. It should also help the reader to distinguish between prescriptive knowledge (the set of design requirements and the prototype) and descriptive knowledge (the conceptual framework and the hierarchical decision model). The fifth node "Teleco case study" is in italics because it is based on the literature and not developed by us in this research. It refers to the idiographic knowledge that we used to validate our prototype. Thus, this article is part of a build and instantiate artifact strategy that proposes an artifact and instantiates it in a prototype.

Let us mention a few limitations. The first one is related to the lack of empirical data supporting our framework. We have validated the usefulness of our prototype, but we still need to evaluate the usefulness of our tool in assisting decision makers in the back sourcing decision [42]. Our system was viewed positively by the practitioners we interviewed in companies. However, they have not yet had the opportunity to implement it. A lesson learned was that the effort of developing tools is really the compilation of the back sourcing literature with theories. However, we need to verify the completeness of the resulting knowledge base. Our

set of design requirements is mainly based on the literature review. It could be enriched with input from practitioners. Companies and organizations are subject to rapid changes in their environment and in the way they operate. This can challenge and partially render obsolete the knowledge established in the publications we have used to feed the conceptual framework of our approach. The hierarchical multicriteria model has the advantages of readability and ease of implementation. However, some rules are likely to resist this purely hierarchical organization. The weights associated with each node are subject to adaptation over time and according to the economic sector. Finally, the current prototype is not yet equipped with an interface adapted to the cognitive style of IS/IT back sourcing decision makers.

A future line of research will examine the approach in a dynamic way, incorporating time. Issues such as the implementation of the back sourcing decision, the evaluation of switching costs, and the management of capabilities should be studied. We will develop metrics on the outcomes of back sourcing decisions to verify the approach's usefulness. Finally, due to the generality of our framework, we will explore other IS/IT problems that can be better addressed by such a framework with adapted features. We believe that the general architecture (the first four levels of the hierarchy) of our approach can be adapted to any management decision that is complex, strategic, and involves a trade-off between different decisions supported by different organizational theories. The criteria level needs to be updated in the current context. Our

approach helps researchers conduct more in-depth analysis for more reliable and realistic results. Our study points in the right direction and can be used as a reference for future research in related areas by both academics and practitioners.

CRediT authorship contribution statement

Jacky Akoka: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Isabelle Comyn-Wattiau:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Appendix

Table A

The set of criteria and associated theories.

Criterion	Possible scenarios	Explanation	Relevant theory
Presented in Table 3			
Strategic			
Socio-organizational Structure			
Organizational structures			
Management change	Renegotiate = 1 Change Vendor = 0 Backsource = 1 Renegotiate = 1 Change Vendor = 1	The strategy development of organizations depends heavily on human factors	Human resource based theory
Structural changes (e.g. new business line, new corporate entity)	Backsource = 1 Renegotiate = 0 Change Vendor = 1	Companies learn through feedback loop and exchange resources across organizational boundaries	Systems Theory
Changes in vendor organization	Change Vendor = 1 Backsource = 1	If the company is unable to strike a balance between the various stakeholders, the company must end the relationship with the current vendor	Stakeholder theory
People			
Increasing dependence with outsourcers	Renegotiate = 0 Change Vendor = 0 Backsource = 1 Renegotiate = 0 Change Vendor = 1	Overdependence on the IT supplier is considered a major outsourcing risk	Resource dependence theory
Poor service quality	Backsource = 1 Renegotiate = 0 Change Vendor = 0	If service quality deteriorates, customers have the ability to terminate the relationship	Social Exchange Theory
Loss of competence	Change Vendor = 0 Backsource = 1 Renegotiate = 0	A core activity should not be outsourced	Theory of core competencies
Internal competences available	Change Vendor = 0 Backsource = 1	Organizational competence is related to economic information that cannot be easily transferred	Competence theory

(continued on next page)

Table A (continued)

Criterion	Possible scenarios	Explanation	Relevant theory
Policies			
IS/IT resource accessibility	Renegotiate = 0 Change Vendor = 0 Backsource = 1 Renegotiate = 0	The organization's ability to integrate, develop and reconfigure internal and external capabilities to cope with rapidly changing environments	Dynamic capabilities Theory
Deterioration of relationship with outsourcer	Change Vendor = 1 Backsource = 1 Renegotiate = 0	The agent (vendor) and the principal (customer) must have common goals and the same level of commitment	Agency theory
Uncertainty on quality control	Change Vendor = 0 Backsource = 1	The outsourcing contract does not easily implement the norms that inform the behavior in the relationship between the company and the vendor	Relational Exchange Theory
Environment External			
External environmental change	Renegotiate = 0 Change Vendor = 0 Backsource = 1 Renegotiate = 0	Mechanisms for monitoring the environment for emerging technologies that may provide strategic advantage are difficult to outsource	Institutional Theory
Legal dispute	Change Vendor = 0 Backsource = 1 Renegotiate = 0	External forces push companies to behave in one way and not behave in another	Institutional Theory
Interest conflict with outsourcing partners	Change Vendor = 1 Backsource = 1 Renegotiate = 0	The company and the supplier must have the same contractual behavior (role integrity, reciprocity, planning, obtaining consent, trust, etc.)	Contractual Theory
Contract problems	Change Vendor = 1 Backsource = 1 Renegotiate = 0	The company and the supplier must have the same contractual behavior (role integrity, reciprocity, planning, obtaining consent, trust, etc.)	Contractual Theory
External bandwagon effect	Change Vendor = 1 Backsource = 1 Renegotiate = 0	Strategy development of organizations is heavily dependent on human factors	Human resource based theory
Breakthrough technology	Change Vendor = 0 Backsource = 1	The outsourcer is not capable of introducing the innovation desired by the client company	Theory of innovation
Internal			
Disclosure of commercial secrets	Renegotiate = 0 Change Vendor = 0 Backsource = 1 Renegotiate = 0	The agent (vendor) and the principal (customer) must have common goals and the same level of commitment	Agency theory
Poor relationship with outsourcer	Change Vendor = 1 Backsource = 1 Renegotiate = 0	A partnership and alliance with the supplier involves interdependence based on trust, cooperation, shared values, relationships interpersonal and regular communication	Partnership and alliance theory
Weakening culture	Change Vendor = 0 Backsource = 1 Renegotiate = 0	The outsourcer is not capable of introducing the innovation desired by the client company	Theory of innovation
Recognition of IT as business enabler	Change Vendor = 0 Backsource = 1 Renegotiate = 0	The main objective of a company is to maximize long-term profits and develop a sustainable competitive advantage over its competitors	Profit maximizing and competition-based theory
Internal bandwagon effect	Change Vendor = 0 Backsource = 1	Strategy development of organizations is heavily dependent on human factors	Human resource-based theory
Economic			
Economic cycles	Renegotiate = 0 Change Vendor = 0 Backsource = 1	Re-internalize activities when transaction costs are high	Transaction Cost Theory

(continued on next page)

Table A (continued)

Criterion	Possible scenarios	Explanation	Relevant theory
Excessive cost	Renegotiate = 0 Change Vendor = 0 Backsource = 1 Renegotiate = 0	Bring operations inside the company when it allows cost reduction	Transaction Cost Theory
Transaction and hidden costs	Change Vendor = 0 Backsource = 1 Renegotiate = 0	Bringing operations inside the company will help identify transaction and hidden costs	Transaction Cost Theory
Outsourcer fails to achieve profit	Change Vendor = 1 Backsource = 1	The main objective of a company is to maximize long-term profits and develop a sustainable competitive advantage over its competitors	Profit maximizing and competition-based theory

Data availability

No data was used for the research described in the article.

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Jacky Akoka is Professor Emeritus at the Conservatoire National des Arts et Métiers (CNAM). He holds a PhD in Applied Mathematics from the University of Paris VI and a PhD in Information Systems from the Massachusetts Institute of Technology (MIT). At the CNAM, he held the Chair of Information Systems. He teaches information systems and decision support systems. His research interests include data and information governance, IS/IT sourcing problems, risk analysis in computer security, conceptual modeling and ontologies, analytics system design, model-driven approaches, and design science research. He has published more than two hundred papers in national and international scientific journals and conferences. His research appeared in journals such as *Journal of Management Information Systems*, *Information and Management*, *Decision Support Systems*, *IEEE Transactions on Computers*, *Data and Knowledge Engineering*, and in the proceedings of major academic conferences including the *International Conference on Conceptual Modeling (ER)*, the *European Conference on Information Systems (ECIS)*, and the *Pacific-Asia Conference on Information Systems (PACIS)*. He is notably consulting editor of the journal *Data and Knowledge Engineering* (Elsevier).

Isabelle Comyn-Wattiau is a professor of information and computer systems at ESSEC Business School (France) where she founded a chair in Information Strategy and Governance. She received an M.S. and a Ph.D. in computer science from the University of Paris Sorbonne. Her research interests include information systems design, information systems governance, conceptual modeling, data and information quality and security, and schema integration. She has published more than one hundred journal and conference papers. Her research has appeared in journals such as *Decision Support Systems*, *Data and Knowledge Engineering*, *Expert Systems with Applications*, *Journal of Management Information Systems*, and major academic conferences including the *International Conference on Conceptual Modeling (ER)* and the *European Conference on Information Systems (ECIS)*. She is notably associate editor of the journal *Data and Knowledge Engineering* (Elsevier).